

Chapter 8

Theory of Constrained Optimization

8.1 Constrained Optimization Problems

A general formulation for nonlinear constrained optimization is

$$\min_{x \in R^n} f(x) \quad (8.1.1)$$

$$\text{s.t.} \quad c_i(x) = 0, \quad i = 1, \dots, m_e; \quad (8.1.2)$$

$$c_i(x) \geq 0, \quad i = m_e + 1, \dots, m, \quad (8.1.3)$$

where the objective function $f(x)$ and the constrained functions $c_i(x)$, ($i = 1, \dots, m$) are all smooth, real-valued functions on R^n , and at least one is nonlinear, and m_e and m are nonnegative integers with $0 \leq m_e \leq m$. Sometimes, we set

$$E = \{1, \dots, m_e\} \text{ and } I = \{m_e + 1, \dots, m\}$$

as index sets of equality constraints and inequality constraints, respectively. If $m = 0$, the problem (8.1.1)-(8.1.3) is an unconstrained optimization problem; if $m_e = m \neq 0$, the problem is called an equality constrained optimization problem; if all $c_i(x)$ ($i = 1, \dots, m$) are linear functions, the problem (8.1.1)-(8.1.3) is called a linearly constrained optimization problem. A linearly constrained optimization problem with quadratic objective function $f(x)$ is said to be a quadratic programming problem which will be discussed in Chapter 9.

Definition 8.1.1 *The point $x \in R^n$ is said to be a feasible point if and only if (8.1.2)-(8.1.3) hold. The set of all feasible points is said to be a feasible set.*

In problem (8.1.1)–(8.1.3), (8.1.2)–(8.1.3) are constrained conditions. From Definition 8.1.1, the feasible point is the point satisfying all constraints. We write the feasible set X as

$$X = \left\{ x \mid \begin{array}{l} c_i(x) = 0, \quad i = 1, \dots, m_e; \\ c_i(x) \geq 0, \quad i = m_e + 1, \dots, m \end{array} \right\}. \quad (8.1.4)$$

or

$$X = \{x \mid c_i(x) = 0, \quad i \in E; \quad c_i(x) \geq 0, \quad i \in I\}. \quad (8.1.5)$$

So, we can rewrite problem (8.1.1)–(8.1.3) as

$$\min_{x \in X} f(x) \quad (8.1.6)$$

which means that solution of constrained optimization problem (8.1.1)–(8.1.3) is just to find a point x on the feasible set X , such that the objective function $f(x)$ is minimized.

In the following, we give some definitions about local and global minimizers.

Definition 8.1.2 *If $x^* \in X$ and if*

$$f(x) \geq f(x^*), \quad \forall x \in X, \quad (8.1.7)$$

then x^ is said to be a global minimizer of the problem (8.1.1)–(8.1.3). If $x^* \in X$ and if*

$$f(x) > f(x^*), \quad \forall x \in X, x \neq x^*, \quad (8.1.8)$$

then x^ is said to be a strict global minimizer.*

Definition 8.1.3 *If $x^* \in X$ and if there is a neighborhood $B(x^*, \delta)$ of x^* such that*

$$f(x) \geq f(x^*), \quad \forall x \in X \cap B(x^*, \delta), \quad (8.1.9)$$

then x^ is said to be a local minimizer of problem (8.1.1)–(8.1.3), where*

$$B(x^*, \delta) = \{x \mid \|x - x^*\|_2 \leq \delta\} \quad (8.1.10)$$

and $\delta > 0$.

If $x^* \in X$ and if there is a neighborhood $B(x^*, \delta)$ of x^* such that

$$f(x) > f(x^*), \quad \forall x \in X \cap B(x^*, \delta), x \neq x^*, \quad (8.1.11)$$

then x^* is said to be a strict local minimizer.

Definition 8.1.4 If $x^* \in X$ and if there is a neighborhood $B(x^*, \delta)$ of x^* such that x^* is the only local minimizer in $X \cap B(x^*, \delta)$, then x^* is an isolated local minimizer.

Obviously, a global minimizer is also a local minimizer.

Assume that x^* is a local minimizer of problem (8.1.1)–(8.1.3), if there is an index $i_0 \in I = [m_e + 1, m]$ such that

$$c_{i_0}(x^*) > 0, \quad (8.1.12)$$

then, if we delete the i_0 -th constraint, x^* is still the local minimizer of the problem obtained by deleting i_0 -th constraint. Thus, we say that the i_0 -th constraint is inactive at x^* . Now, we give the definitions of active constraint and inactive constraint. First, write

$$I(x) = \{i \mid c_i(x) = 0, i \in I\}. \quad (8.1.13)$$

Definition 8.1.5 For any $x \in R^n$, the set

$$\mathcal{A}(x) = E \cup I(x) \quad (8.1.14)$$

is an index set of active constraints at x , $c_i(x)$ ($i \in \mathcal{A}(x)$) is an active constraint at x , $c_i(x)$ ($i \notin \mathcal{A}$) is an inactive constraint at x .

Assume that $\mathcal{A}(x^*)$ is an index set of the active constraints of problem (8.1.1)–(8.1.3) at x^* , then, from the observation about inactive constraints, it is enough for us to solve the constrained optimization problem

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & c_i(x) = 0, \quad i \in \mathcal{A}(x^*). \end{aligned} \quad (8.1.15)$$

In general, it is easier to solve equality constraint problem (8.1.15) than the original problem (8.1.1)–(8.1.3).

8.2 First-Order Optimality Conditions

In this section we discuss the first-order optimality conditions. Since the feasible directions play a very important role in deriving the optimality conditions, we first give some definitions of several feasible directions.

Definition 8.2.1 Let $x^* \in X$, $0 \neq d \in R^n$. If there exists $\delta > 0$ such that

$$x^* + td \in X, \quad \forall t \in [0, \delta],$$

then d is said to be a feasible direction of X at x^* . The set of all feasible directions of X at x^* is

$$FD(x^*, X) = \{d \mid x^* + td \in X, \quad \forall t \in [0, \delta]\}. \quad (8.2.1)$$

Definition 8.2.2 Let $x^* \in X$ and $d \in R^n$. If

$$\begin{aligned} d^T \nabla c_i(x^*) &= 0, & i \in E, \\ d^T \nabla c_i(x^*) &\geq 0, & i \in I(x^*), \end{aligned}$$

then d is said to be a linearized feasible direction of X at x^* . The set of all linearized feasible directions of X at x^* is

$$LFD(x^*, X) = \left\{ d \mid \begin{array}{l} d^T \nabla c_i(x^*) = 0, \quad i \in E \\ d^T \nabla c_i(x^*) \geq 0, \quad i \in I(x^*) \end{array} \right\}. \quad (8.2.2)$$

Definition 8.2.3 Let $x^* \in X$ and $d \in R^n$. If there exist sequences $d_k (k = 1, 2, \dots)$ and $\delta_k > 0, (k = 1, 2, \dots)$ such that $x^* + \delta_k d_k \in X, \quad \forall k$ and $d_k \rightarrow d, \delta_k \rightarrow 0$, then the limiting direction d is called the sequential feasible direction of X at x^* . The set of all sequential feasible directions of X at x^* is

$$SFD(x^*, X) = \left\{ d \mid \begin{array}{l} x^* + \delta_k d_k \in X, \quad \forall k \\ d_k \rightarrow d, \delta_k \rightarrow 0 \end{array} \right\}. \quad (8.2.3)$$

In the definition above, if set $x_k = x^* + \delta_k d_k$, then $\{x_k\}$ is a feasible point sequence that satisfies:

- (1) $x_k \neq x^*, \forall k$;
- (2) $\lim_{k \rightarrow \infty} x_k = x^*$;

(3) $x_k \in X$ for all k sufficiently large.

If set $\delta_k = \|x_k - x^*\|$, then we have

$$d_k = \frac{x_k - x^*}{\|x_k - x^*\|} \rightarrow d,$$

which means that $x_k = x^* + \delta_k d_k$ is a feasible point sequence with the feasible direction d .

Note that if $SFD(x^*, X)$ includes the zero vector, it is referred to as the tangent cone of X at x^* , i.e.,

$$\mathcal{T}_X(x^*) = SFD(x^*, X) \cup \{0\}.$$

Obviously, by use of the above definitions of some feasible directions, we have the following lemma which indicates the relations of the above sets of feasible directions $FD(x^*, X)$, $SFD(x^*, X)$ and $LFD(x^*, X)$.

Lemma 8.2.4 *Let $x^* \in X$. If all constraint functions are differentiable at x^* , then*

$$FD(x^*, X) \subseteq SFD(x^*, X) \subseteq LFD(x^*, X). \quad (8.2.4)$$

Proof. For any $d \in FD(x^*, X)$, it follows from Definition 8.2.1 that there is a $\delta > 0$ such that (8.2.1) holds. Set $d_k = d$ and $\delta_k = \delta/2^k$, then (8.2.3) holds and clearly $d_k \rightarrow d$ and $\delta_k \rightarrow 0$. Thus $d \in SFD(x^*, X)$. Since d is arbitrary, then

$$FD(x^*, X) \subseteq SFD(x^*, X). \quad (8.2.5)$$

Next, for any $d \in SFD(x^*, X)$, if $d = 0$, then $d \in LFD(x^*, X)$. Assume that $d \neq 0$. By Definition 8.2.3, there exist sequences d_k ($k = 1, 2, \dots$) and $\delta_k > 0$ ($k = 1, 2, \dots$) such that (8.2.3) holds, and $d_k \rightarrow d \neq 0$ and $\delta_k \rightarrow 0$. By use of (8.2.3), we see that $x^* + \delta_k d_k \in X$, i.e.,

$$0 = c_i(x^* + \delta_k d_k) = \delta_k d_k^T \nabla c_i(x^*) + o(\|\delta_k d_k\|), \quad i \in E; \quad (8.2.6)$$

$$0 \leq c_i(x^* + \delta_k d_k) = \delta_k d_k^T \nabla c_i(x^*) + o(\|\delta_k d_k\|), \quad i \in I(x^*). \quad (8.2.7)$$

Dividing the above two equations by $\delta_k > 0$ and setting $k \rightarrow \infty$, we obtain (8.2.2). Thus we also have

$$SFD(x^*, X) \subseteq LFD(x^*, X). \quad (8.2.8)$$

Both (8.2.5) and (8.2.8) give the result of (8.2.4). $\square$

In order to describe clearly necessary conditions for a local solution, it is convenient to introduce the set

$$\mathcal{D}(x') = \mathcal{D}' = \{d \mid d^T \nabla f(x') < 0\}, \quad (8.2.9)$$

which is called a set of descent direction at x' .

Now we describe the most basic necessary condition – geometry optimality condition as follows.

Theorem 8.2.5 (*Geometry optimality condition*) *Let $x^* \in X$ be a local minimizer of problem (8.1.1)-(8.1.3). If $f(x)$ and $c_i(x)$ ($i = 1, 2, \dots, m$) are differentiable at x^* , then*

$$d^T \nabla f(x^*) \geq 0, \quad \forall d \in SFD(x^*, X), \quad (8.2.10)$$

which means

$$SFD(x^*, X) \cap \mathcal{D}(x^*) = \phi, \quad (8.2.11)$$

where ϕ is an empty set.

Proof. For any $d \in SFD(x^*, X)$, there exist $\delta_k > 0$ ($k = 1, 2, \dots$) and d_k ($k = 1, 2, \dots$) such that $x^* + \delta_k d_k \in X$ with $\delta_k \rightarrow 0$ and $d_k \rightarrow d$. Since $x^* + \delta_k d_k \rightarrow x^*$ and x^* is a local minimizer, then for k sufficiently large, we have

$$f(x^*) \leq f(x^* + \delta_k d_k) = f(x^*) + \delta_k d_k^T \nabla f(x^*) + o(\delta_k) \quad (8.2.12)$$

which implies

$$d^T \nabla f(x^*) \geq 0. \quad (8.2.13)$$

Since d is arbitrary, we obtain (8.2.10).

Furthermore, (8.2.13) also implies $d \notin \mathcal{D}(x^*)$, and hence $SFD(x^*, X) \cap \mathcal{D}(x^*) = \phi$. $\square$

If we use terminology of the tangent cone to represent (8.2.10), we have

$$d^T \nabla f(x^*) \geq 0, \quad \forall d \in \mathcal{T}_X(x^*),$$

i.e.,

$$-\nabla f(x^*)^T d \leq 0, \quad \forall d \in \mathcal{T}_X(x^*). \quad (8.2.14)$$

This implies that

$$-\nabla f(x^*) \in \mathcal{N}_X(x^*), \quad (8.2.15)$$

where $\mathcal{N}_X(x^*)$ is the normal cone of X at x^* .

Theorem 8.2.5 shows that there is no sequential feasible direction at a local minimizer x^* . Unfortunately, it is not possible to proceed further without constraint qualification. In the following, by means of Farkas' Lemma 1.3.22 and the constraint qualification, we can get the first-order optimality condition — the famous Karush-Kuhn-Tucker Theorem.

Farkas' Lemma 1.3.22 gives the following form.

Lemma 8.2.6 *The set*

$$S = \left\{ d \left| \begin{array}{l} d^T \nabla f(x^*) < 0, \\ d^T \nabla c_i(x^*) = 0, \ i \in E, \\ d^T \nabla c_i(x^*) \geq 0, \ i \in I \end{array} \right. \right\} \quad (8.2.16)$$

is empty if and only if there exist real numbers $\lambda_i, i \in E$ and nonnegative real numbers $\lambda_i \geq 0, i \in I$ such that

$$\nabla f(x^*) = \sum_{i \in E} \lambda_i \nabla c_i(x^*) + \sum_{i \in I} \lambda_i \nabla c_i(x^*). \quad (8.2.17)$$

In fact, set

$$d = -x, \nabla f(x^*) = c, A = \begin{bmatrix} \nabla c_1^T(x^*) \\ \vdots \\ \nabla c_m^T(x^*) \end{bmatrix}, \lambda = y,$$

we immediately have that (8.2.16) is just (1.3.49), and that (8.2.17) and $\lambda_i \geq 0, i \in I$ are just (1.3.50). This implies that Lemma 8.2.6 is a direct consequence of Farkas' Lemma 1.3.22 and also called Farkas' Lemma.

It is convenient to state the optimality condition by introducing the Lagrangian function

$$\mathcal{L}(x, \lambda) = f(x) - \sum_{i=1}^m \lambda_i c_i(x), \quad (8.2.18)$$

where $\lambda = (\lambda_1, \dots, \lambda_m)^T \in R^m$ is a Lagrange multiplier vector.

Now, we are in a position to state the first-order necessary condition of a local minimizer by use of Farkas' Lemma and Theorem 8.2.5.

Theorem 8.2.7 (*Karush-Kuhn-Tucker Theorem*)

Let x^* be a local minimizer of problem (8.1.1)–(8.1.3). If the constraint qualification (CQ)

$$SFD(x^*, X) = LFD(x^*, X) \quad (8.2.19)$$

holds, then there exist Lagrange multipliers λ_i^* such that the following conditions are satisfied at (x^*, λ^*) :

$$\nabla f(x^*) - \sum_{i=1}^m \lambda_i^* \nabla c_i(x^*) = 0, \quad (8.2.20)$$

$$c_i(x^*) = 0, \quad \forall i \in E, \quad (8.2.21)$$

$$c_i(x^*) \geq 0, \quad \forall i \in I, \quad (8.2.22)$$

$$\lambda_i^* \geq 0, \quad \forall i \in I, \quad (8.2.23)$$

$$\lambda_i^* c_i(x^*) = 0, \quad \forall i \in I. \quad (8.2.24)$$

Proof. Since x^* is a local minimizer, x^* is feasible and the conditions (8.2.21) and (8.2.22) are satisfied.

Let $d \in SFD(x^*, X)$; since x^* is a local minimizer, it follows from Theorem 8.2.5 that $d^T \nabla f(x^*) \geq 0$. By constraint qualification (8.2.19), we have $d \in LFD(x^*, X)$. Thus the system

$$d^T \nabla c_i(x^*) = 0, \quad i \in E, \quad (8.2.25)$$

$$d^T \nabla c_i(x^*) \geq 0, \quad i \in I(x^*), \quad (8.2.26)$$

$$d^T \nabla f(x^*) < 0 \quad (8.2.27)$$

has no solution. By Farkas' Lemma, we immediately obtain that

$$\nabla f(x^*) = \sum_{i \in E} \lambda_i^* \nabla c_i(x^*) + \sum_{i \in I(x^*)} \lambda_i^* \nabla c_i(x^*), \quad (8.2.28)$$

where $\lambda_i^* \in R$ ($i \in E$) and $\lambda_i^* \geq 0$ ($i \in I(x^*)$). Setting $\lambda_i^* = 0$ ($i \in I \setminus I(x^*)$), it follows that

$$\nabla f(x^*) = \sum_{i=1}^m \lambda_i^* \nabla c_i(x^*),$$

which is (8.2.20). It is obvious that $\lambda_i^* \geq 0, \forall i \in I$.

Finally, note that:

when $i \in I(x^*)$, $c_i(x^*) = 0$ and $\lambda_i^* \geq 0$, therefore $\lambda_i^* c_i(x^*) = 0$;

when $i \in I \setminus I(x^*)$, $c_i(x^*) > 0$ but $\lambda_i^* = 0$, therefore we also have $\lambda_i^* c_i(x^*) = 0$.

Thus we obtain that $\lambda_i^* c_i(x^*) = 0, \forall i \in I$. $\square$

Theorem 8.2.7 was presented by Kuhn and Tucker [193], and is known as the Kuhn-Tucker Theorem. Since Karush [185] also considered similarly the optimality condition for constrained optimization, the conditions (8.2.20)–(8.2.24) are often known as the Karush-Kuhn-Tucker conditions, or KKT conditions for short. A point that satisfies the conditions is referred to as a KKT point.

In KKT conditions, (8.2.20) is called a stationary point condition, because it can be rewritten as

$$\nabla_x \mathcal{L}(x^*, \lambda^*) = \nabla f(x^*) - \sum_{i=1}^m \lambda_i^* \nabla c_i(x^*) = 0. \quad (8.2.29)$$

Conditions (8.2.21) and (8.2.22) are called the feasibility conditions, (8.2.23) is the nonnegativity condition for multipliers, and (8.2.24) is referred to as the complementarity condition which states that both λ_i^* and $c_i(x^*)$ cannot be nonzero, or equivalently that Lagrange multipliers corresponding to inactive constraints are zero.

We say that the strict complementarity condition holds if exactly one of λ_i^* and $c_i(x^*)$ is zero for each $i \in I$, i.e., we have that $\lambda_i^* > 0$ for each $i \in I \cap \mathcal{A}(x^*)$.

An inequality constraint c_i is strongly active if $i \in I \cap \mathcal{A}(x^*)$ and $\lambda_i^* > 0$, i.e., $\lambda_i^* > 0$ and $c_i(x^*) = 0$. An inequality constraint c_i is weakly active if $i \in I \cap \mathcal{A}(x^*)$ and $\lambda_i^* = 0$, i.e., $\lambda_i^* = c_i(x^*) = 0$.

The condition (8.2.19) is called the constraint qualification (CQ). The constraint qualification is important for KKT conditions. As an example given by Fletcher [133], it indicates that if constraint qualification (8.2.19) does not hold, then the local minimizer of problem (8.1.1)–(8.1.3) may not be a KKT point.

Example:

$$\min_{(x_1, x_2) \in \mathbb{R}^2} \quad x_1 \quad (8.2.30)$$

$$\text{s.t.} \quad x_1^3 - x_2 \geq 0, \quad (8.2.31)$$

$$x_2 \geq 0. \quad (8.2.32)$$

It is not difficult to see that $x^* = (0, 0)^T$ is the global minimizer of (8.2.30)–(8.2.32). At x^* , we have

$$SFD(x^*, X) = \left\{ d \left| d = \begin{pmatrix} \alpha \\ 0 \end{pmatrix}, \alpha \geq 0 \right. \right\} \quad (8.2.33)$$

and

$$LFD(x^*, X) = \left\{ d \mid d = \begin{pmatrix} \alpha \\ 0 \end{pmatrix}, \alpha \in R^1 \right\}. \quad (8.2.34)$$

Therefore, (8.2.19) does not hold. By direct computing, we have

$$\nabla f(x^*) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \nabla c_1(x^*) = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \quad \nabla c_2(x^*) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad (8.2.35)$$

which show that there does not exist λ_1^* and λ_2^* such that

$$\nabla f(x^*) = \lambda_1^* \nabla c_1(x^*) + \lambda_2^* \nabla c_2(x^*). \quad (8.2.36)$$

This simple example indicates the importance of constraint qualification. However, it is not easy to examine whether or not the CQ condition (8.2.19) holds. In the following, we give some concrete constraint qualifications which are easy to examine and frequently used.

The most simple and obvious constraint qualification is linear function constraint qualification.

Definition 8.2.8 *If all constraints $c_i(x^*)$ ($i \in \mathcal{A}(x^*) = E \cup I(x^*)$) are linear functions, we say that linear function constraint qualification (LFCQ) holds.*

By the definition, if $c_i(x^*)$ ($i \in \mathcal{A}(x^*)$) are linear functions, then CQ condition (8.2.19) holds and we have the following corollary.

Corollary 8.2.9 *Let x^* be a local minimizer of problem (8.1.1)-(8.1.3). If the linear function constraint qualification holds at x^* , then x^* is a KKT point.*

The most important and frequently used constraint qualification is the following linear independence constraint qualification (LICQ).

Definition 8.2.10 *If active constraint gradients $\nabla c_i(x^*)$, $i \in \mathcal{A}(x^*)$ are linearly independent, we say that the linear independence constraint qualification (LICQ) holds.*

Theorem 8.2.11 *Let x^* be a feasible point and $\mathcal{A}(x^*)$ an index set of active constraints at x^* . If $\nabla c_i(x^*)$, $i \in \mathcal{A}(x^*)$, are linearly independent, then the constraint qualification (8.2.19) holds.*

Proof. Since $SFD(x^*, X) \subseteq LFD(x^*, X)$, it is enough that we only need to prove $LFD(x^*, X) \subseteq SFD(x^*, X)$. Let $d \in LFD(x^*, X)$ be arbitrary. Now let

$$\mathcal{A}(x^*) = E \cup I(x^*) = \{1, \dots, l\}, \quad m_e \leq l \leq n.$$

Since $\nabla c_1(x^*), \dots, \nabla c_l(x^*)$ are linearly independent, there are $b_{l+1}, \dots, b_n$ such that $\nabla c_1(x^*), \dots, \nabla c_l(x^*), b_{l+1}, \dots, b_n$ are linearly independent.

Consider the nonlinear system

$$r(x, \theta) = 0, \quad (8.2.37)$$

whose components are defined as

$$r_i(x, \theta) = c_i(x) - \theta d^T \nabla c_i(x^*), \quad i = 1, \dots, l, \quad (8.2.38)$$

$$r_i(x, \theta) = (x - x^*)^T b_i - \theta d^T b_i, \quad i = l + 1, \dots, n. \quad (8.2.39)$$

When $\theta = 0$, the system (8.2.37) is solved by x^* , and when $\theta \geq 0$ is sufficiently small, any solution x is also a feasible point in (8.1.1)–(8.1.3).

Let us write

$$A = [\nabla c_1(x), \dots, \nabla c_l(x)], \quad B = [b_{l+1}, \dots, b_n].$$

Then the Jacobian matrix $J(x, \theta) = \nabla_x r^T(x, \theta) = [A : B]$. Obviously, $J(x^*) = [A(x^*) : B]$ is nonsingular. Hence by the implicit function theorem there exist open neighborhoods Ω_x about x^* and Ω_θ about $\theta = 0$ such that for any $\theta \in \Omega_\theta$, a unique solution $x(\theta) \in \Omega_x$ exists, and $x(\theta)$ is feasible and continuously differentiable with respect to θ . From (8.2.37) and using the chain rule,

$$0 = \frac{dr_i}{d\theta} = \sum_j \frac{\partial r_i}{\partial x_j} \frac{dx_j}{d\theta} + \frac{\partial r_i}{\partial \theta}, \quad i = 1, \dots, n,$$

that is

$$\nabla c_i(x)^T \frac{dx}{d\theta} - \nabla c_i(x^*)^T d = 0, \quad i = 1, \dots, l, \quad (8.2.40)$$

$$b_i^T \frac{dx}{d\theta} - b_i^T d = 0, \quad i = l + 1, \dots, n. \quad (8.2.41)$$

The above system is

$$J^T \frac{dx}{d\theta} - J(x^*)^T d = 0.$$

Since $x = x^*$ at $\theta = 0$, we have $J = J(x^*)$ at $\theta = 0$. Thus the above equation becomes

$$J(x^*)\left[\frac{dx}{d\theta}\bigg|_{\theta=0} - d\right] = 0.$$

Since the coefficient matrix is nonsingular, we obtain

$$\frac{dx}{d\theta} = d \text{ at } \theta = 0,$$

which implies that if $\theta_k \downarrow 0$ is any sequence, then $x(\theta_k)$ is a feasible sequence with the feasible direction d , i.e.,

$$\frac{x(\theta_k) - x^*}{\theta_k} \rightarrow d.$$

This shows that $d \in SFD(x^*, X)$. Since $d \in LFD(x^*, X)$ is arbitrary, we get $LFD(x^*, X) \subseteq SFD(x^*, X)$. $\square$

By the above theorem and Theorem 8.2.7, we immediately obtain the following theorem.

Theorem 8.2.12 *Let x^* be a local minimizer of problem (8.1.1)–(8.1.3). If LICQ holds, i.e., $\nabla c_i(x^*), i \in \mathcal{A}(x^*) = E \cup I(x^*)$, are linearly independent, then there are Lagrange multipliers λ_i^* ($i = 1, \dots, m$) such that (8.2.20)–(8.2.24) hold.*

We want to mention that sometimes we use the regularity assumption

$$SFD(x^*, X) \cap \mathcal{D}(x^*) = LFD(x^*, X) \cap \mathcal{D}(x^*). \quad (8.2.42)$$

Since both sides are subsets of $SFD(x^*, X)$ and $LFD(x^*, X)$ respectively, this assumption is clearly implied by the CQ (8.2.19). However, the converse does not hold.

With the regularity assumption (8.2.42), the necessary condition (8.2.14) in Theorem 8.2.5 (no feasible descent directions: $SFD(x^*, X) \cap \mathcal{D}(x^*) = \phi$) becomes

$$LFD(x^*, X) \cap \mathcal{D}(x^*) = \phi,$$

i.e., there are no linearized feasible descent directions. Furthermore, as a corollary of KKT Theorem 8.2.7, we have

Theorem 8.2.13 *Let x^* be a local minimizer of problem (8.1.1)–(8.1.3). If the regularity assumption (8.2.42) holds, then x^* is a KKT point.*

Next, we discuss the first-order sufficiency condition.

Theorem 8.2.14 *Let $x^* \in X$. Let $f(x)$ and $c_i(x)$ ($i = 1, \dots, m$) be differentiable at x^* . If*

$$d^T \nabla f(x^*) > 0, \quad \forall 0 \neq d \in SFD(x^*, X), \quad (8.2.43)$$

then x^ is a strict local minimizer of problem (8.1.1)–(8.1.3).*

Proof. Suppose, by contradiction, that x^* is not a strict local minimizer, then there exist $x_k \in X$ ($k = 1, 2, \dots$) such that

$$f(x_k) \leq f(x^*), \quad (8.2.44)$$

and $x_k \rightarrow x^*, x_k \neq x^*$ ($k = 1, 2, \dots$). Without loss of generality, we assume that

$$\frac{x_k - x^*}{\|x_k - x^*\|_2} \rightarrow d. \quad (8.2.45)$$

Set $d_k = (x_k - x^*)/\|x_k - x^*\|_2, \delta_k = \|x_k - x^*\|_2$. By Definition 8.2.3, we have

$$d \in SFD(x^*, X). \quad (8.2.46)$$

By use of (8.2.44), (8.2.45) and $f(x_k) = f(x^*) + (x_k - x^*)^T \nabla f(x^*) + o(\|x_k - x^*\|_2)$, by dividing $\|x_k - x^*\|_2$ and then taking the limit as $k \rightarrow \infty$, we obtain

$$d^T \nabla f(x^*) \leq 0, \quad (8.2.47)$$

which, together with (8.2.46), contradicts (8.2.43). This completes the proof. $\square$

Since $SFD(x^*, X) \subseteq LFD(x^*, X)$, we also have the following corollary.

Corollary 8.2.15 *Let $x^* \in X$. Let $f(x)$ and $c_i(x)$ ($i = 1, \dots, m$) be differentiable at x^* . If*

$$d^T \nabla f(x^*) > 0, \quad \forall 0 \neq d \in LFD(x^*, X), \quad (8.2.48)$$

then x^ is a strict local minimizer of problem (8.1.1)–(8.1.3).*

The other important optimality condition, which is credited to Fritz John [183], is the Fritz John optimality condition, which needs no the constraint qualification.

Theorem 8.2.16 *Let $f(x)$ and $c_i(x)$ ($i = 1, \dots, m$) be continuously differentiable on a nonempty open set containing the feasible set X . If x^* is a local minimizer of problem (8.1.1)–(8.1.3), then there exist a scalar $\lambda_0^* \geq 0$ and a vector $\lambda^* \in R^m$ such that*

$$\lambda_0^* \nabla f(x^*) - \sum_{i=1}^m \lambda_i^* \nabla c_i(x^*) = 0, \quad (8.2.49)$$

$$c_i(x^*) = 0, \quad i \in E, \quad (8.2.50)$$

$$c_i(x^*) \geq 0, \quad i \in I, \quad (8.2.51)$$

$$\lambda_i^* \geq 0, \quad i \in I, \quad (8.2.52)$$

$$\lambda_i^* c_i(x^*) = 0, \quad \forall i, \quad (8.2.53)$$

$$\sum_{i=0}^m (\lambda_i^*)^2 > 0. \quad (8.2.54)$$

Proof. If $\nabla c_i(x^*)$ ($i \in \mathcal{A}(x^*)$) are linearly dependent, then there are λ_i^* ($i \in \mathcal{A}(x^*)$) not all zero, such that

$$\sum_{i \in \mathcal{A}(x^*)} \lambda_i^* \nabla c_i(x^*) = 0. \quad (8.2.55)$$

Set $\lambda_0^* = 0$ and $\lambda_i^* = 0$, ($i \in I \setminus \mathcal{A}(x^*)$), we obtain (8.2.49)–(8.2.54).

If $\nabla c_i(x^*)$ ($i \in \mathcal{A}(x^*)$) are linearly independent, we can obtain immediately (8.2.49)–(8.2.54) with $\lambda_0 = 1$ by means of Theorem 8.2.12. $\square$

The point satisfying (8.2.49)–(8.2.54) is said to be the Fritz John point. The following weighted Lagrangian function

$$\tilde{L}(x, \lambda_0, \lambda) = \lambda_0 f(x) - \sum_{i=1}^m \lambda_i c_i(x) \quad (8.2.56)$$

is said to be the Fritz John function. Obviously, the Fritz John point is the stationary point of the Fritz John function. Note that $\lambda_0 \geq 0$. If $\lambda_0 > 0$, the Fritz John function can be regarded as a λ_0 multiple of the Lagrangian function. However, if $\lambda_0 = 0$, the Fritz John function only describes the constraint functions and is independent of the objective function. In such a case, Fritz John optimality conditions do not represent actually the optimality conditions of the original constrained optimization problem. This disadvantage makes the Fritz John conditions unfavorable.

We conclude this section with an optimality condition of convex programming.

As we know, the problem of minimizing a convex function on a convex set Ω is said to be a convex programming problem. Such a problem has the form

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & x \in \Omega, \end{aligned} \tag{8.2.57}$$

where $f(x)$ is a convex function on a convex set Ω . Typically, in nonlinear programming

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & c_i(x) = 0, \ i \in E, \\ & c_i(x) \geq 0, \ i \in I, \end{aligned} \tag{8.2.58}$$

if $f(x)$ is convex, $c_i(x)$, ($i \in E$) are linear functions, and $c_i(x)$, ($i \in I$) are concave, then the constrained set $\Omega = \{x \mid c_i(x) = 0, i \in E; c_i(x) \geq 0, i \in I\}$ is a convex set, and hence the problem (8.2.58) is convex programming.

As Theorem 1.4.9 in the unconstrained case, the following theorem indicates that the local minimizer of convex programming is also its global minimizer.

Theorem 8.2.17 *Each local minimizer of convex programming problem (8.2.57) is also the global minimizer, and the set S of global minimizers is convex.*

Proof. Suppose, by contradiction, that x^* is a local but not global minimizer. Then there exists $x_1 \in \Omega$ such that $f(x_1) < f(x^*)$. Consider

$$x_\theta = (1 - \theta)x^* + \theta x_1, \ \theta \in [0, 1].$$

By convexity of Ω , $x_\theta \in \Omega$. Also, by convexity of f ,

$$\begin{aligned} f(x_\theta) &\leq (1 - \theta)f(x^*) + \theta f(x_1) \\ &= f(x^*) + \theta(f(x_1) - f(x^*)) \\ &< f(x^*). \end{aligned}$$

For sufficiently small θ , $x_\theta \in N(x^*, \epsilon) \cap \Omega$. So, it follows from assumption that x^* is a local minimizer that $f(x_\theta) \geq f(x^*)$. We get a contradiction which means that local minimizers are global.

Let $x_0, x_1 \in S$. Define $x_\theta = (1 - \theta)x_0 + \theta x_1, \theta \in [0, 1]$. By the global property, $f(x_\theta) \geq f(x_0) = f(x_1)$. However, by convexity of f , $f(x_\theta) \leq (1 - \theta)f(x_0) + \theta f(x_1) = f(x_0) = f(x_1)$. Therefore $f(x_\theta) = f(x_0) = f(x_1)$ and so $x_\theta \in S$, which means that S is convex. $\square$

Theorem 8.2.18 *The KKT point of convex programming must be its minimizer.*

Proof. Let (x^*, λ^*) be any KKT pair of convex programming. Obviously, the Lagrangian function

$$L(x, \lambda^*) = f(x) - \sum_{i \in E} \lambda_i^* c_i(x) - \sum_{i \in I} \lambda_i^* c_i(x) \quad (8.2.59)$$

is convex for x . By use of properties of convex function and KKT conditions, we have for any feasible x ,

$$\begin{aligned} L(x, \lambda^*) &\geq L(x^*, \lambda^*) + (x - x^*)^T \nabla_x L(x^*, \lambda^*) \\ &= L(x^*, \lambda^*) \\ &= f(x^*) - \sum_{i=1}^m \lambda_i^* c_i(x^*) \\ &= f(x^*). \end{aligned} \quad (8.2.60)$$

Note that x is a feasible point and $\lambda_i^* \geq 0, i \in I$, so we have

$$\lambda_i^* c_i(x) = 0, i \in E; \lambda_i^* c_i(x) \geq 0, i \in I.$$

Hence

$$L(x, \lambda^*) \leq f(x). \quad (8.2.61)$$

By (8.2.60) and (8.2.61) we obtain

$$f(x) \geq f(x^*), \quad (8.2.62)$$

that is, KKT point x^* is a minimizer. $\square$

Theorem 8.2.19 *The convex programming with strictly convex objective function has unique minimizer.*

The proof of this theorem is as an exercise.

8.3 Second-Order Optimality Conditions

We have seen in unconstrained optimization that the second-order derivative information has significant implications in optimality conditions. Let $x^* \in X$. If

$$d^T \nabla f(x^*) > 0, \forall 0 \neq d \in SFD(x^*, X), \quad (8.3.1)$$

then x^* is a strict local minimizer of problem (8.1.1)–(8.1.3). If

$$\text{there exists } d \in SFD(x^*, X) \text{ such that } d^T \nabla f(x^*) < 0, \quad (8.3.2)$$

then from Theorem 8.2.5 it follows that x^* must not be a local minimizer of problem (8.1.1)–(8.1.3). These results tell us that, provided either (8.3.1) or (8.3.2) holds, the first-order optimality condition can be used to determine whether x^* is a local minimizer. However, we cannot determine whether x^* is a local minimizer by the first derivative information alone, if both (8.3.1) and (8.3.2) do not hold, i.e.,

$$d^T \nabla f(x^*) \geq 0, \forall d \in SFD(x^*, X); \quad (8.3.3)$$

$$d^T \nabla f(x^*) = 0, \exists 0 \neq d \in SFD(x^*, X). \quad (8.3.4)$$

In these cases, the second-order derivative information is needed.

Assume that the constraint qualification (8.2.19) holds. It follows from (8.3.3), (8.2.19) and Farkas' Lemma 8.2.6 that x^* is a KKT point. By (8.3.4) and the definition of Lagrange multipliers, there exists $0 \neq d \in SFD(x^*, X)$ such that

$$d^T \nabla f(x^*) = \sum_{i=1}^m \lambda_i^* d^T \nabla c_i(x^*) = 0. \quad (8.3.5)$$

Since $SFD(x^*, X) \subseteq LFD(x^*, X)$, by use of Definition 8.2.2, we have that (8.3.5) is equivalent to

$$\lambda_i^* d^T \nabla c_i(x^*) = 0, \forall i \in I(x^*). \quad (8.3.6)$$

So, we give the following definitions. Let x^* be a KKT point of (8.1.1)–(8.1.3), and λ^* a corresponding Lagrange multiplier vector. Define a set of strong active constraints as

$$I_+(x^*) = \{i \mid i \in I(x^*) \text{ with } \lambda_i^* > 0\}. \quad (8.3.7)$$

Obviously, $I_+(x^*) \subseteq I(x^*)$.

Definition 8.3.1 Let x^* be a KKT point of (8.1.1)–(8.1.3), and λ^* a corresponding Lagrange multiplier vector. If there exist sequences d_k ($k = 1, 2, \dots$) and $\delta_k > 0$ ($k = 1, 2, \dots$) such that

$$x^* + \delta_k d_k \in X \quad (8.3.8)$$

satisfy

$$c_i(x_k) = 0, \quad i \in E \cup I_+(x^*), \quad (8.3.9)$$

$$c_i(x_k) \geq 0, \quad i \in I(x^*) \setminus I_+(x^*), \quad (8.3.10)$$

and $d_k \rightarrow d$ and $\delta_k \rightarrow 0$, then d is said to be a sequential null constraint direction at x^* . The set of all sequential null constraint directions is written as $S(x^*, \lambda^*)$,

$$S(x^*, \lambda^*) = \left\{ d \mid \begin{array}{l} x_k = x^* + \delta_k d_k \in X, \delta_k > 0, \delta_k \rightarrow 0, d_k \rightarrow d, \\ c_i(x_k) = 0, \quad i \in E \cup I_+(x^*), \\ c_i(x_k) \geq 0, \quad i \in I(x^*) - I_+(x^*). \end{array} \right\}. \quad (8.3.11)$$

It is easy to see that (8.3.9)–(8.3.10) imply that

$$\sum_{i=1}^m \lambda_i^* c_i(x^* + \delta_k d_k) = 0. \quad (8.3.12)$$

So, equivalently,

$$S(x^*, \lambda^*) = \left\{ d \mid \begin{array}{l} d \in SFD(x^*, X); \\ \sum_{i=1}^m \lambda_i^* c_i(x_k) = 0. \end{array} \right\}. \quad (8.3.13)$$

Obviously, $S(x^*, \lambda^*) \subseteq SFD(x^*, X)$.

Similar to the linearized feasible direction, we have the following definition.

Definition 8.3.2 Let x^* be a KKT point of (8.1.1)–(8.1.3), and λ^* a corresponding Lagrange multiplier vector. If d is a linearized feasible direction at x^* and (8.3.6) holds, then d is said to be a linearized null constraint direction. The set of all linearized null constraint directions is written as $G(x^*, \lambda^*)$,

$$G(x^*, \lambda^*) = \left\{ d \mid \begin{array}{l} d \neq 0, \\ d^T \nabla c_i(x^*) = 0, \quad i \in E \cup I_+(x^*), \\ d^T \nabla c_i(x^*) \geq 0, \quad i \in I(x^*) \setminus I_+(x^*). \end{array} \right\}. \quad (8.3.14)$$

Equivalently,

$$G(x^*, \lambda^*) = \left\{ d \mid \begin{array}{l} d \in LFD(x^*, \lambda^*); \\ d^T \nabla c_i(x^*) = 0, i \in I_+(x^*). \end{array} \right\} \quad (8.3.15)$$

If the Lagrange multiplier at x^* is unique, $G(x^*, \lambda^*)$ can be denoted by $G(x^*)$.

By the above definitions, we have

$$S(x^*, \lambda^*) \subseteq SFD(x^*, X), \quad (8.3.16)$$

$$G(x^*, \lambda^*) \subseteq LFD(x^*, X). \quad (8.3.17)$$

Similar to $SFD(x^*, X) \subseteq LFD(x^*, X)$, we also can prove

$$S(x^*, \lambda^*) \subseteq G(x^*, \lambda^*), \quad (8.3.18)$$

which is an exercise left to readers.

Now, we state the main results of this section.

Theorem 8.3.3 (Second-order necessary conditions)

Let x^* be a local minimizer of (8.1.1)–(8.1.3). If the constraint qualification (8.2.19) holds, then we have

$$d^T \nabla_{xx}^2 \mathcal{L}(x^*, \lambda^*) d \geq 0, \quad \forall d \in S(x^*, \lambda^*), \quad (8.3.19)$$

where $\mathcal{L}(x, \lambda)$ is a Lagrangian function.

Furthermore, if

$$S(x^*, \lambda^*) = G(x^*, \lambda^*), \quad (8.3.20)$$

then

$$d^T \nabla_{xx}^2 \mathcal{L}(x^*, \lambda^*) d \geq 0, \quad \forall d \in G(x^*, \lambda^*). \quad (8.3.21)$$

Proof. For any $d \in S(x^*, \lambda^*)$, if $d = 0$, it is obvious that $d^T \nabla_{xx}^2 \mathcal{L}(x^*, \lambda^*) d = 0$. Now we consider $d \neq 0$. From the definition of $S(x^*, \lambda^*)$, there exist $\{d_k\}$ and $\{\delta_k\}$ such that (8.3.8)–(8.3.12) hold. Therefore, by (8.3.12) and KKT conditions, we have

$$\begin{aligned} f(x^* + \delta_k d_k) &= \mathcal{L}(x^* + \delta_k d_k, \lambda^*) \\ &= \mathcal{L}(x^*, \lambda^*) + \frac{1}{2} \delta_k^2 d_k^T \nabla_{xx}^2 \mathcal{L}(x^*, \lambda^*) d_k + o(\delta_k^2) \\ &= f(x^*) + \frac{1}{2} \delta_k^2 d_k^T \nabla_{xx}^2 \mathcal{L}(x^*, \lambda^*) d_k + o(\delta_k^2). \end{aligned} \quad (8.3.22)$$

Since x^* is a local minimizer, it follows for all k sufficiently large that

$$f(x^* + \delta_k d_k) \geq f(x^*). \quad (8.3.23)$$

Using (8.3.22)–(8.3.23) and taking limits give

$$d^T \nabla_{xx}^2 \mathcal{L}(x^*, \lambda^*) d \geq 0.$$

Since $d \in S(x^*, \lambda^*)$ is arbitrary, then (8.3.19) follows.

By (8.3.20), we immediately obtain (8.3.21) from (8.3.19). $\square$

Theorem 8.3.4 (*Second-order sufficient conditions*)

Let x^ be a KKT point of (8.1.1)–(8.1.3). If*

$$d^T \nabla_{xx}^2 \mathcal{L}(x^*, \lambda^*) d > 0, \quad \forall d \in G(x^*, \lambda^*), \quad (8.3.24)$$

then x^ is a strict local minimizer.*

Proof. Assume that x^* is not a strict local minimizer, then there exists a sequence $\{x_k\} \subset X$ such that

$$f(x_k) \leq f(x^*), \quad (8.3.25)$$

with $x_k \rightarrow x^*$ and $x_k \neq x^*$ ($k = 1, 2, \dots$). Without loss of generality, we assume that

$$\frac{x_k - x^*}{\|x_k - x^*\|_2} \rightarrow d.$$

By a similar argument to (8.2.45)–(8.2.47), we have

$$d^T \nabla f(x^*) \leq 0 \quad (8.3.26)$$

and

$$d \in SFD(x^*, X) \subseteq LFD(x^*, X). \quad (8.3.27)$$

It follows by KKT conditions and (8.2.4) that

$$d^T \nabla f(x^*) = \sum_{i=1}^m \lambda_i d^T \nabla c_i(x^*) \geq 0. \quad (8.3.28)$$

Note that (8.3.26) and (8.3.28) give

$$d^T \nabla f(x^*) = 0 \quad (8.3.29)$$

which implies from (8.3.28) and Definition 8.2.2 that

$$\lambda_i d^T \nabla c_i(x^*) = 0, \quad \forall i \in I(x^*). \quad (8.3.30)$$

So, it follows from (8.3.27), (8.3.30) and Definition 8.3.2 that

$$d \in G(x^*, \lambda^*). \quad (8.3.31)$$

From (8.3.25), we have

$$\begin{aligned} \mathcal{L}(x^*, \lambda^*) &\geq \mathcal{L}(x_k, \lambda^*) \\ &= \mathcal{L}(x^*, \lambda^*) + \frac{1}{2} \delta_k^2 d_k^T \nabla_{xx}^2 \mathcal{L}(x^*, \lambda^*) d_k + o(\delta_k^2). \end{aligned} \quad (8.3.32)$$

Dividing by δ_k^2 and taking the limit give

$$d^T \nabla_{xx}^2 \mathcal{L}(x^*, \lambda^*) d \leq 0 \quad (8.3.33)$$

which contradicts (8.3.24). We complete the proof. $\square$

Notice that a sufficient condition for (8.3.24) is that

$$d^T \nabla_{xx}^2 \mathcal{L}(x^*, \lambda^*) d > 0$$

for all $d \neq 0$ such that $d^T \nabla c_i(x^*) = 0$, $i \in \mathcal{A}_+(x^*, \lambda^*)$, where

$$\mathcal{A}_+(x^*, \lambda^*) = E \cup \{i \mid i \in I(x^*), \lambda_i^* > 0\}, \quad (8.3.34)$$

which is obtained by deleting indices for which $\lambda_i^* = 0$, $i \in I(x^*)$ from $\mathcal{A}(x^*)$. The $\mathcal{A}_+(x^*, \lambda^*)$ is said to be an index set of strong active constraints, which is a union of the index sets of equality constraints and strongly active inequality constraints. So, we immediately obtain the following corollary which is also a second-order sufficient condition and more convenient to verify in practice.

Corollary 8.3.5 *Let x^* be a KKT point of (8.1.1)–(8.1.3). If*

$$d^T \nabla_{xx}^2 \mathcal{L}(x^*, d^*) d > 0 \quad (8.3.35)$$

for all d satisfying

$$d^T \nabla c_i(x^*) = 0, \quad \forall i \in \mathcal{A}_+(x^*, \lambda^*), \quad (8.3.36)$$

then x^ is a strict local minimizer.*

Proof. It is enough to prove that (8.3.35)–(8.3.36) are the sufficient conditions of (8.3.24). In fact, if x^* is a KKT point, then $\forall d \in SFD(x^*, X) \subseteq LFD(x^*, X)$,

$$d^T \nabla c_i(x^*) = 0, \quad i \in E \quad (8.3.37)$$

$$d^T \nabla c_i(x^*) \geq 0, \quad i \in I(x^*). \quad (8.3.38)$$

So, (8.3.6) holds, which implies $d \in G(x^*, \lambda^*)$ from Definition 8.3.2. Therefore, by means of Theorem 8.3.4, it follows that (8.3.35)–(8.3.36) implies (8.3.24). $\square$

8.4 Duality

We conclude this chapter with a brief discussion of duality. The concept of duality occurs widely in the mathematical programming literature. The aim is to provide an alternative formulation of a mathematical programming problem which is more convenient computationally or has some theoretical significance.

The original problem is referred to as the primal, and the transformed problem is referred to as the dual.

In this section, we give an introduction of duality theory which is associated with the convex programming problem. We will introduce the Lagrangian dual problem, and prove the duality theorem and the weak duality theorem. Now we first state the duality theorem.

Theorem 8.4.1 *Let x^* be a minimizer of convex primal problem (P)*

$$\begin{aligned} \min_x \quad & f(x) \\ \text{s.t.} \quad & c_i(x) \geq 0, \quad i = 1, \dots, m. \end{aligned} \quad (8.4.1)$$

If $f(x)$ and $c_i(x)$, ($i = 1, \dots, m$) are continuously differentiable and the regularity condition (8.2.42) holds, then x^ and λ^* solve the dual problem*

$$\begin{aligned} \max_{x, \lambda} \quad & \mathcal{L}(x, \lambda) \\ \text{s.t.} \quad & \nabla_x \mathcal{L}(x, \lambda) = 0, \\ & \lambda \geq 0. \end{aligned} \quad (8.4.2)$$

Furthermore, the minimum of the primal and the maximum of the dual are equal, i.e.,

$$f(x^*) = \mathcal{L}(x^*, \lambda^*). \quad (8.4.3)$$

Proof. By the assumption and KKT Theorem 8.2.7, there exist Lagrange multipliers $\lambda^* \geq 0$ such that $\nabla_x \mathcal{L}(x^*, \lambda^*) = 0$ and $\lambda_i^* c_i(x^*) = 0$, $i = 1, \dots, m$. Thus, $f(x^*) = \mathcal{L}(x^*, \lambda^*)$.

Let x, λ be dual feasible. Using $\lambda \geq 0$, convexity of $\mathcal{L}$, and $\nabla_x \mathcal{L}(x, \lambda) = 0$ gives

$$\begin{aligned}
 \mathcal{L}(x^*, \lambda^*) &= f(x^*) \geq f(x^*) - \sum_{i=1}^m \lambda_i^* c_i(x^*) \\
 &= \mathcal{L}(x^*, \lambda) \\
 &\geq \mathcal{L}(x, \lambda) + (x^* - x)^T \nabla_x \mathcal{L}(x, \lambda) \\
 &= \mathcal{L}(x, \lambda)
 \end{aligned} \tag{8.4.4}$$

which means that (x^*, λ^*) solves the dual problem. $\square$

Usually, (8.4.3) is said to be the strong duality. Now, we give some examples of dual problems. Let the primal problem in linear programming be

$$\begin{aligned}
 \min \quad & c^T x \\
 \text{s.t.} \quad & Ax \geq b.
 \end{aligned} \tag{8.4.5}$$

By Theorem 8.4.1, we immediately have the dual:

$$\begin{aligned}
 \max \quad & b^T \lambda \\
 \text{s.t.} \quad & A\lambda = c, \\
 & \lambda \geq 0.
 \end{aligned} \tag{8.4.6}$$

Normally, linear programs have standard form:

$$\begin{aligned}
 \min \quad & c^T x \\
 \text{s.t.} \quad & Ax = b, \\
 & x \geq 0.
 \end{aligned} \tag{8.4.7}$$

The corresponding dual problem is

$$\begin{aligned}
 \max \quad & b^T \lambda \\
 \text{s.t.} \quad & A^T \lambda \leq c.
 \end{aligned} \tag{8.4.8}$$

It is easy to examine that the optimality conditions of (8.4.7) and (8.4.8) are identical.

For convex quadratic programming, the primal problem is

$$\begin{aligned} \min_x \quad & \frac{1}{2}x^T Gx + h^T x \\ \text{s.t.} \quad & A^T x \geq b, \end{aligned} \tag{8.4.9}$$

where G is positive definite. The dual problem is

$$\begin{aligned} \max_{x, \lambda} \quad & \frac{1}{2}x^T Gx + h^T x - \lambda^T (A^T x - b) \\ \text{s.t.} \quad & Gx + h - A\lambda = 0, \end{aligned} \tag{8.4.10}$$

$$\lambda \geq 0. \tag{8.4.11}$$

By eliminating x , we obtain the following problem:

$$\begin{aligned} \max_{\lambda} \quad & -\frac{1}{2}\lambda^T (A^T G^{-1} A)\lambda + \lambda^T (b - A^T G^{-1} h) - \frac{1}{2}h^T G^{-1} h \\ \text{s.t.} \quad & \lambda \geq 0. \end{aligned} \tag{8.4.12}$$

This is a quadratic programming problem in λ with bounded-constraints $\lambda \geq 0$.

The following theorem, referred to as the weak duality theorem, shows that the objective value of any feasible point of the primal problem is larger than or equal to the objective value of any feasible point of the dual problem.

Theorem 8.4.2 *Let x' be any feasible point in primal problem (8.4.1). Let (x, λ) be any feasible point in dual problem (8.4.2). Then*

$$f(x') \geq \mathcal{L}(x, \lambda). \tag{8.4.13}$$

Proof. Let x' be primal feasible and (x, λ) dual feasible. Then by convexity of f , dual feasibility, concavity of c_i , and nonnegativity of $c_i(x')$ and λ_i in turn, it follows that

$$\begin{aligned} f(x') - f(x) & \geq \nabla f(x)^T (x' - x) \\ & = \sum_{i=1}^m \lambda_i \nabla c_i(x)^T (x' - x) \\ & \geq \sum_{i=1}^m \lambda_i (c_i(x') - c_i(x)) \\ & \geq -\sum_{i=1}^m \lambda_i c_i(x). \end{aligned}$$

Hence

$$f(x') \geq f(x) - \sum_{i=1}^m \lambda_i c_i(x) = \mathcal{L}(x, \lambda). \quad \square$$

From the above theorem, we immediately have

$$\inf_x f(x) \geq \sup_{x, \lambda} \mathcal{L}(x, \lambda). \quad (8.4.14)$$

This implies that if the primal problem is unbounded, it follows that $\inf_x f(x) = \sup_{x, \lambda} \mathcal{L}(x, \lambda) = -\infty$, and this is not possible if (x, λ) is feasible. Therefore, an unbounded primal implies an inconsistent dual.

Exercises

1. Assume that $f(x)$ is a convex function, $c_i(x)$ ($1 \leq i \leq m_e$) are linear functions and $c_i(x)$ ($m_e + 1 \leq i \leq m$) are concave functions. Show that x^* is a global minimizer of (8.1.1)–(8.1.3) if it is a local minimizer of (8.1.1)–(8.1.3).

2. Define the ϵ -active set by

$$I_\epsilon(x) = \{i \mid c_i(x) \leq \epsilon, \quad i \in I(x)\}.$$

Prove that, for any given $x \in \Re^n$,

$$\lim_{\epsilon \rightarrow 0_+} I_\epsilon(x) = I(x).$$

3. Prove: if $c_i(x)$ ($i \in \mathcal{A}(x^*)$) are linear functions, then CQ condition (8.2.19) holds.

4. Prove (8.3.16) and (8.3.17).

5. Prove (8.3.18).

6. Let $0 \neq c \in \Re^n$. Consider the problem

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & \|x\|_2^2 \leq 1. \end{aligned}$$

Prove that $x^* = c/\|c\|_2$ satisfies the second-order sufficient condition.

7. Form the KKT conditions for

$$\begin{array}{ll} \max & (x+1)^2 + (y+1)^2 \\ \text{s.t.} & x^2 + y^2 \leq 2, \\ & 1 - y \geq 0 \end{array}$$

and then determine the solution.

8. Give an example in which the second-order necessary condition holds while the second-order sufficient condition fails.

9. By solving the KKT equation, find the point on the ellipse defined by the intersection of the surface $x + y = 1$ and $x^2 + 2y^2 + z^2 = 1$ which is nearest to the origin.

10. Show that the dual of problem

$$\begin{array}{ll} \min & \frac{1}{2}\sigma x_1^2 + \frac{1}{2}x_2^2 + x_1 \\ \text{s.t.} & x_1 \geq 0 \end{array}$$

is a maximization problem in terms of a Lagrange multiplier λ . For the case $\sigma = +1$ and $\sigma = -1$, investigate whether the local solution of the dual gives the multiplier λ^* which exists at the local solution to the primal.